

TITLE OF THE COURSE				
Course Code	22AD42		Credits	3
Hours/Week (L-T-P-S)	3-0-0		CIE Marks	50
Total Teaching Hours	39		SEE Marks	50
Exam Hours	03		Course Type	Theory
Course Component				
COURSE LEARNING OUTCOMES				
Students will be able to:				
1. Describe the evolution of artificial intelligence along with search methods and applications				
2. Apply the principles of search strategies to relevant problems				
3. To design algorithms to find optimal paths and apply concepts of A* algorithm in developing applications				
4. Describe and apply the principles and steps of Game Playing Algorithms				
5. To apply first order logic and describe the usage of Prolog in realization of Knowledge base for querying and describe concepts of planning and GRAPHPLAN algorithm				
COURSE CONTENTS				
UNIT -1 (08 Hrs)				
HISTORY OF AI AND STATE SPACE SEARCH: History, Can Machines think? Turing Test, Winograd Schema Challenge, Language, Philosophy, Mind, Reasoning, Computation, The Chess Saga, Intelligent Agents State Space Search: Depth First Search, Breadth First Search, Depth First Iterative Deepening , Heuristic Search: Best First Search, Hill Climbing, Solution Space, Travelling Salesman Problem				
UNIT -2 (08 Hrs)				
SEARCH STRATEGIES AND ALGORITHMS: Escaping Local Optima, Random Walk Algorithm, introduction to Simulated Annealing, Population based methods, Concepts of Genetic Algorithms and application to Eight Queens problem, SAT, TSP, Emergent Systems, Overview of Concepts of Ant Colony Optimization and application to TSP ; Finding Optimal Paths: Branch & Bound, A*, Admissibility of A*, Informed Heuristic Functions				
UNIT -3 (07 Hrs)				
MORE ON A* ALGORITHM AND GAME PLAYING: Results related to Admissibility of A* , Space Saving Version of A*: Iterative Deepening A*, Game Playing: Board Games and Game Trees, Algorithm MiniMax, Algorithm AlphaBeta. Limitations of Search; overview of examples, Chess, Go and Backgammon				
UNIT -4 (07 Hrs)				
FIRST ORDER LOGIC AND PROLOG: Rule Based Expert Systems, Knowledge, Reasoning and Planning, First order logic, Knowledge base; Application: A Logic based Financial Advisor; Concepts of Programming using Prolog				
UNIT -5 (09 Hrs)				
PLANNING: Planning Domain Definition Language and examples such as Air Cargo Transport, Block World, The complexity of classical planning, Forward(progression) state space search Backward(regression) relevant- states search, GRAPHPLAN algorithm, Means Ends Analysis				

TEXTBOOKS					
SINO	Unit	Textbook Title	Author(s)	Publisher(s)	Edition/Year of Publication
1.	All	A First Course in Artificial Intelligence	Deepak Khemani	McGraw Hill Education (India)	2013
REFERENCE BOOKS					
1	All	Artificial Intelligence Structures and Strategies for Complex Problem Solving	George F Luger	Pearson	6th Edition

ONLINE RESOURCES	
Topic/Title	Link
An introduction to Artificial Intelligence	https://onlinecourses.nptel.ac.in/noc24_cs08
Artificial Intelligence: Knowledge Representation and Reasoning	https://onlinecourses.nptel.ac.in/noc24_cs14
COURSE ASSESSMENT METHOD	
Continuous Internal Evaluation (CIE): Course Project/Assignment -20 Marks Three internals, 30 Marks each will be conducted and the Average of all three will be taken. Semester End Examination (SEE): Final examination of 100 Marks will be conducted and will be evaluated for 50 Marks.	
PEDAGOGY	

CO-PO-PSO MAPPING														
CO	PO												PSO1	PSO2
	1	2	3	4	5	6	7	8	9	10	11	12		
1	03													03
2	03	02				02							02	03
3	03	02				02							02	03
4	03	02					02						02	03
5	03	02					02						02	03